audio.dev
CONFERENCE
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Meteora

A reverb computed from a real landscape

Trent Eriksen

THE IDEA

What if a reverb's space were a place you move through?

Meteora places your sound among the rock pillars and monasteries of Meteora, in Thessaly, Greece. Two microphones move through the valley, and every cliff answers at its real distance and direction, through air that changes with the weather.

Simulated from open data · uncalibrated



THE MODEL

From open data to an echo

Open data
Copernicus 30 m elevation; OpenStreetMap cliffs and summits

→

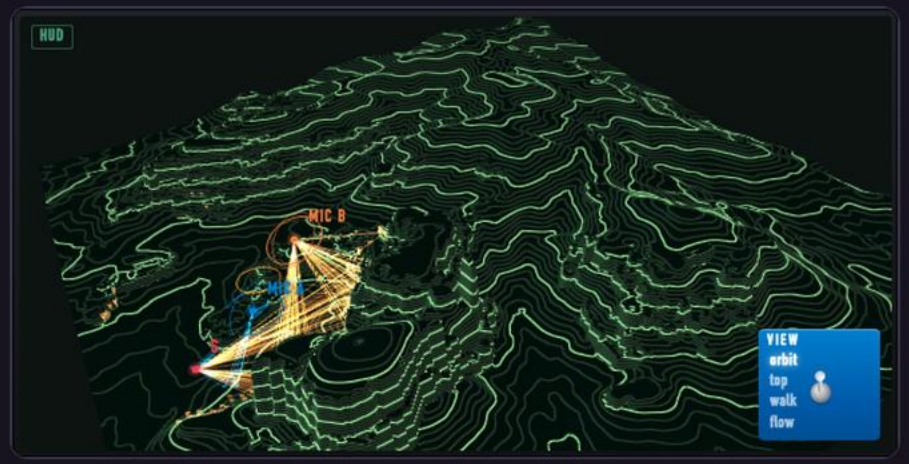
Site model
A 5 m terrain; pillars sharpened, every change logged

→

Site pack
57,000 surface patches and named rock faces

→

Echo field
First order echoes per microphone; a traced, convolved tail



An offline Python pipeline builds the landscape once. In the plugin, echoes are traced again on a worker whenever a microphone moves, and the audio thread only plays delay taps and the tail.

THE PHYSICS

Distance, direction and air are physical

1.75 s
for a cliff 300 m away to answer

57,000
acoustic surface patches on a 5 m terrain

2 mics
anywhere in the valley, with modelled polar patterns

ISO 9613-1
air absorption with temperature, humidity and wind



The echo scope: every echo the two microphones hear, and the tail. Loudness and tone are plausible estimates, not measurements.

THE INSTRUMENT

A panel after the Buchla 200 series



Site model

The terrain as contour lines, the microphones and the echo paths, live

Echo scope

Every echo and the tail, as they change

Modules

Placement, air, microphones, distance, wind, faces, shape

Patch row

Random, function, listener and oscillator sources into any control

LEARN mode

Explains every control on the panel

NEW IN 0.3

A reverb that listens

attacks · density · brightness · pitch · noisiness

OFF | **LISTEN** | **SETTINGS** | **PATCH** | **ALIVE**

From suggestions only to a place with a life of its own, subtle to drastic

- Suggests pot values and cables, or moves them itself
- Never writes your host automation: LOCK, KEEP, A/B and RETURN keep you in charge
- The EXTREME lamp lights whenever it leaves the physics



IN MOTION

Plugin preview



Meteora 0.3 in use in Ableton Live: a 45 second excerpt. Click the frame to play it here.

Full video on YouTube: youtube.com/watch?v=vMqgGGazfks

IN MOTION

Artistic preview



Meteora 0.3, edited with photographs by the author: the instrument and the place it comes from, together (1 min). Click the frame to play it here.

Watch it on YouTube:
youtube.com/watch?v=8rZie27Bz_8

RELEASED

Tested, then released free for ADC

Check	Result
pluginval, strictness 10	pass
Steinberg VST3 validator	47 of 47
Address, thread and realtime sanitizers	no reports
Panel soak tests under live use	pass
CPU, 48 kHz, 128 samples, M1 Max	about 12 % of a core



Free download

macOS, Apple Silicon
VST3 and Standalone
github.com/Grashopr-888/meteora-releases

NEXT

Toward Meteora 1.0

1.0 for musicians

A notarised installer with no Terminal step, an AU for Logic, one stable plugin identity, a simpler first screen

More platforms

Windows (VST3) and the CLAP format

A better ear

The listener and the guide tuned on real players and recordings



Thank you · Trent Eriksen

grashopr-888.github.io/current-projects/projects/meteora · github.com/Grashopr-888/meteora-releases

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